***Neuroanatomy*: Blood Supply of Brain & Spinal Cord**

1. The major arterial supply to the brain is divided into two systems. Which artery enters the cranium through the carotid canal to form the anterior circulation system?
a. Vertebral
b. Basilar
c. Maxillary
d. Superficial temporal
e. Internal carotid
2. The two vertebral arteries ascend through the foramina transversaria of the cervical vertebrae and enter the skull through which anatomical opening?
a. Foramen ovale
b. Foramen rotundum
c. Foramen spinosum
d. Foramen magnum
e. Jugular foramen
3. Inside the cranium, the two vertebral arteries ascend on the ventral surface of the medulla oblongata and fuse at the lower border of the pons to form a single midline vessel called the:
a. Anterior spinal artery
b. Posterior cerebral artery
c. Internal carotid artery
d. Basilar artery
e. Middle cerebral artery
4. The Circle of Willis provides crucial collateral circulation at the base of the brain. Which specific artery connects the internal carotid system to the posterior cerebral artery?
a. Anterior communicating
b. Middle cerebral
c. Anterior cerebral
d. Ophthalmic
e. Posterior communicating
5. Anastomosis between the right and left anterior cerebral arteries across the midline of the brain is established directly by which short vascular channel?
a. Posterior communicating
b. Internal carotid
c. Basilar
d. Anterior communicating
e. Vertebral
6. The primary arterial trunk that passes laterally into the lateral cerebral sulcus and supplies the vast majority of the lateral surface of the cerebral hemisphere is the:
a. Anterior cerebral artery
b. Posterior cerebral artery
c. Middle cerebral artery
d. Anterior choroidal artery
e. Labyrinthine artery

7. A neuroanatomist maps the blood supply to the medial surface of the cerebral cortex, extending from the frontal pole to the parieto-occipital sulcus. This area is supplied by the:
a. Middle cerebral artery
b. Posterior communicating artery
c. Basilar artery
d. Anterior cerebral artery
e. Anterior choroidal artery
8. The visual cortex, located on the occipital lobe surrounding the calcarine sulcus, receives its primary arterial blood supply from which terminal branch of the basilar system?
a. Anterior cerebral artery
b. Middle cerebral artery
c. Anterior choroidal artery
d. Superior cerebellar artery
e. Posterior cerebral artery
9. The striate block of deep penetrating arteries, which branch off the middle cerebral artery to supply the internal capsule and basal ganglia, are called:
a. Cortical branches
b. Medullary branches
c. Spinal branches
d. Thalamoperforating branches
e. Lenticulostriate arteries
10. One of the long, slender lenticulostriate branches of the middle cerebral artery is highly susceptible to rupture in hypertensive patients, causing hemiplegia. This specific vessel is Charcot's artery of:
a. Thrombosis
b. Embolism
c. Aneurysm
d. Hemorrhage
e. Vasospasm
11. The primary somatic motor cortex area controlling the lower limb and foot (paracentral lobule) is located on the medial surface of the hemisphere. It is supplied by the:
a. Anterior cerebral artery
b. Middle cerebral artery
c. Posterior cerebral artery
d. Anterior choroidal artery
e. Posterior communicating artery
12. The specialized collection of vascular capillaries responsible for secreting cerebrospinal fluid into the lateral ventricles is supplied directly by the:
a. Posterior communicating artery
b. Anterior choroidal artery
c. Labyrinthine artery
d. Pontine arteries
e. Anterior spinal artery
13. The superior surface of the cerebellum and the underlying deep cerebellar nuclei receive their primary blood supply from which direct branch of the basilar artery?

a. Posterior inferior cerebellar artery
b. Anterior inferior cerebellar artery
c. Posterior cerebral artery
d. Superior cerebellar artery
e. Anterior spinal artery

14. The posterior inferior cerebellar artery (PICA), which supplies the posterolateral aspect of the medulla oblongata and inferior cerebellar surface, branches directly off the:

a. Basilar artery
b. Internal carotid artery
c. Middle cerebral artery
d. Vertebral artery
e. Posterior cerebral artery

15. The internal auditory meatus transmits a small, delicate branch of the anterior inferior cerebellar artery (or basilar artery) that accompanies the vestibulocochlear nerve. This is the:

a. Superior cerebellar artery
b. Posterior cerebral artery
c. Anterior spinal artery
d. Ophthalmic artery
e. Labyrinthine artery

16. The anterior two-thirds of the spinal cord gray and white matter matter is supplied by a single longitudinal vessel formed by branches of the vertebral arteries. This vessel is the:

a. Posterior spinal artery
b. Anterior spinal artery
c. Great radicular artery
d. Posterior intercostal artery
e. Vertebral canal trunk

17. The posterior one-third of the spinal cord gray and white matter, including the dorsal columns handling proprioception, is supplied by a pair of narrow longitudinal channels called the:

a. Anterior spinal arteries
b. Vertebral arteries
c. Posterior spinal arteries
d. Segmental medullary arteries
e. Intercostal arteries

18. Because the longitudinal spinal arteries are too small to sustain the lower segments of the spinal cord, they are reinforced by spinal inputs called:

a. Cortical branches
b. Basal branches
c. Lenticulostriate vessels
d. Segmental medullary (radicular) arteries
e. Circle of Willis channels

19. A exceptionally large radicular artery branches off an intercostal or lumbar artery to provide the major blood supply to the lower thoracic and lumbar segments of the spinal cord. This is the artery of:

a. Charcot

b. Willis
c. Adamkiewicz (Great radicular artery)
d. Monro
e. Sylvius

20. The deep venous drainage of the internal structures of the brain, including the basal ganglia and internal capsule, consolidates to form the:
a. Superior sagittal sinus
b. Inferior sagittal sinus
c. Great cerebral vein (of Galen)
d. Cavernous sinus
e. Transverse sinus

21. The great cerebral vein of Galen unites with the inferior sagittal sinus within the interior of the cranium to form which dural venous sinus?
a. Superior sagittal sinus
b. Transverse sinus
c. Sigmoid sinus
d. Straight sinus
e. Cavernous sinus

22. Blood from the straight sinus, superior sagittal sinus, and occipital sinus drains into an expansion located internal to the internal occipital protuberance called the:
a. Cavernous sinus
b. Confluence of sinuses
c. Sigmoid sinus
d. Superior petrosal sinus
e. Inferior petrosal sinus

23. The dural venous sinuses exit the cranium to form the internal jugular vein by traveling directly through which lateral skull opening?
a. Foramen magnum
b. Foramen lacerum
c. Jugular foramen
d. Foramen spinosum
e. Hypoglossal canal

24. The specialized dural venous sinus located on either side of the sphenoid bone body, containing the internal carotid artery and cranial nerves III, IV, V, and VI, is the:
a. Superior sagittal sinus
b. Transverse sinus
c. Sigmoid sinus
d. Cavernous sinus
e. Straight sinus

25. Cerebrospinal fluid is absorbed from the subarachnoid space back into the venous system of the superior sagittal sinus through specialized microscopic structures called:
a. Choroid plexuses
b. Ependymal tanycytes
c. Arachnoid granulations (villi)
d. Perivascular Virchow-Robin spaces
e. Dermal pacchionian bodies

26. The localized dilation located at the bifurcation of the common carotid artery, containing specialized baroreceptors that monitor systemic blood pressure, is the:
a. Carotid body
b. Carotid sinus
c. Jugular bulb
d. Subclavian glomus
e. Aortic body
27. The chemoreceptors that monitor blood oxygen concentrations, carbon dioxide tension, and pH are clustered in a small structure at the carotid bifurcation called the:
a. Carotid sinus
b. Jugular ganglion
c. Ciliary body
d. Carotid body
e. Otic ganglion
28. The spinal veins drain sequentially from the spinal cord plexus into the internal vertebral venous plexus, which is located inside which anatomical compartment?
a. Subarachnoid space
b. Subdural space
c. Epidural (extradural) space
d. Central canal
e. Subpial matrix
29. The blood-brain barrier, which dynamically controls the passage of substances from the systemic circulation into the neural tissue, is structurally formed by capillary endothelial tight junctions and:
a. Oligodendrocyte sheets
b. Microglial processes
c. Ependymal linings
d. Astrocyte foot processes (perivascular feet)
e. Pia mater extensions
30. Congenital sac-like dilations that form at the arterial bifurcations of the Circle of Willis due to structural defects in the tunica media are clinically known as:
a. Fusiform aneurysms
b. Dissecting aneurysms
c. Berry (saccular) aneurysms
d. Arteriovenous malformations
e. Varicose dilations
Clinical Touch BCQs: Vascular Syndromes & Pathologies
31. A 68-year-old hypertensive patient is admitted following a severe stroke. Neurological exam reveals contralateral hemiplegia and hemianesthesia that affect the face and upper limb far more severely than the lower limb. Which artery is occluded?
a. Anterior cerebral
b. Posterior cerebral
c. Middle cerebral
d. Vertebral
e. Anterior spinal

32. A patient presents with contralateral weakness and sensory loss confined exclusively to the left lower limb and foot, accompanied by urinary incontinence and a lack of volition (abulia). This presentation points to an occlusion of the:
a. Right middle cerebral artery
b. Right anterior cerebral artery
c. Left posterior cerebral artery
d. Basilar artery
e. Right posterior communicating artery
33. An angiogram reveals a complete occlusion of the posterior cerebral artery. Which clinical sign would you expect to find on an exam due to ischemia of the primary visual cortex?
a. Contralateral homonymous hemianopia with macular sparing
b. Total blindness in both eyes
c. Ipsilateral monocular blindness
d. Bitemporal heteronymous hemianopia
e. Quadrantanopia with macular degeneration
34. A 55-year-old male is diagnosed with Lateral Medullary (Wallenberg) Syndrome. Neurological testing shows loss of pain/temperature on the ipsilateral face and contralateral body, vertigo, and dysphagia. This stroke is caused by occlusion of the:
a. Anterior spinal artery
b. Posterior inferior cerebellar artery (PICA)
c. Middle cerebral artery
d. Anterior choroidal artery
e. Superior cerebellar artery
35. A patient presents with Medial Medullary Syndrome, exhibiting contralateral hemiplegia of the trunk and limbs, contralateral loss of proprioception/vibration, and ipsilateral paralysis with deviation of the tongue. The artery involved is the:
a. Posterior spinal
b. Posterior inferior cerebellar
c. Labyrinthine
d. Anterior spinal artery
e. Middle cerebral
36. A patient experiences sudden quadriplegia, bilateral cranial nerve palsies, and a total loss of voluntary movement, except for vertical eye movements and blinking (Locked-in Syndrome). The vascular occlusion is located in the:
a. Middle cerebral artery
b. Anterior cerebral artery
c. Basilar artery
d. Internal carotid artery
e. Great cerebral vein
37. A 32-year-old female presents with an explosive, excruciating headache ("the worst headache of my life") followed by a loss of consciousness. Lumbar puncture shows blood in the CSF. This acute clinical picture indicates a rupture of a berry aneurysm into the:
a. Epidural space
b. Subdural space
c. Subarachnoid space
d. Lateral ventricles

e. Central canal

38. An elderly male matching a history of transient ischemic attacks (TIAs) suffers a stroke. Imaging confirms ischemia along the sensory and motor fibers traveling through the posterior limb of the internal capsule. This tells us there is an occlusion of the:

a. Anterior cerebral artery cortical branches
b. Lenticulostriate branches of the middle cerebral artery
c. Posterior inferior cerebellar artery
d. Vertebral muscular branches
e. Spinal medullary arteries

39. During an abdominal aortic aneurysm repair, the major segmental radicular inputs to the lower spinal cord are compressed for an extended period. Postoperatively, the patient is paraplegic with loss of pain/temperature below the lesion, but proprioception and vibration remain completely intact. This indicates infarction of the:

a. Dorsal columns
b. Posterior spinal arteries
c. Anterior spinal artery territory
d. Cauda equina
e. Filum terminale matrix

40. A patient presents with severe proptosis (bulging eye), ophthalmoplegia (paralysis of eye movements), and loss of sensation across the forehead following a deep facial infection. This indicates septic thrombosis of the:

a. Superior sagittal sinus
b. Cavernous sinus
c. Straight sinus
d. Transverse sinus
e. Inferior sagittal sinus